

# C++ Programming Exam - Answer Sheet

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## Question 1 (15 points)

1.1 Which line of the above code is both a definition and an initialization?

Statement c

1.2 Which line of the above code declares a variable without allocating memory for it?

Statement b

1.3 Which lines of the above code are assignments?

Statement d and e

1.4 According to the following statement: `const int *p;`

the value of the pointer p cannot change

1.5 When is the following logical test true? `( x>=y && ! x && x* y < 0 && y==0)`

never

1.6 Which of the following cases of mixed expressions is/are correct?

'f' - 'd' is a char

1.7 Given: `char c='b'; char *pc=&c; char *&rc=pc;` then after `(*rc)++;` what happens?

it increases pc, by one byte

1.8 Which of the provided statements (if any) are invalid?

`px = px = rx;`

1.9 Which of the provided statements would give the value of x?

`**ppx, *((double*)pp)`

1.10 Which expressions give as result an int equal to 6?

`'z' - 't', 13 % 7`

1.11 What will be the value of x after: `int x = (7>6 ? 1+8 : 8)`

8

1.12 Which of the following(s) is a (are) valid function declaration?

`void func(int x);`

1.13 Which function will be called by: `printFun(2.0*f)` given float f;

`void printFun(float)`

1.14 How many times is function fib called when num is 3?

3

1.15 In a function with return type void, what happens at return?

0 is returned

## Question 2 (5 points)

2.1  $77 + c + i + 1L$

int

2.2  $6.55f + f / 1.5 - 9 / 8$

double

2.3  $'z' - 'z'$

int

2.4  $b + c$

int

2.5  $'t' - 'a' + c$

char

2.6  $77.8f + 4 * 0.5f + 45L$

float

2.7  $42L + (int) d + 94.3f + int(4.9)$

double

2.8  $0.0 + f + c$

double

2.9  $5.28L * d * 3 + 4.5$

double

2.10  $1.5f / d * f + 6.9 * 4L$

double

## Question 3 (10 points)

Replace the comments with expressions:

```
// Local name: cout << "\n Local name : " << name; // Global name: cout << "\n Global  
name : " << ::name; // Object member name: cout << "\n Object name : " << m.name; //  
Invoke member function print(): m.print();
```

## Question 4 (10 points)

4.1 Fragment 1 output

x is 10 \*y is 10 \*\*z is 10

4.2 Fragment 2 output

a is 1 b is 1 c is 1 x is 1 y is 0 z is 1

4.3 Fragment 3 output

x[0] = 2 y[0] = 1

4.4 Fragment 4 output

test1=20test2=50.8test3=68

4.5 Fragment 5 output

At the beginning a = 10 b = 20 Before func2 a = 5 b = 40 After func2 a = 5 b = 40 At the end a = 10 b = 20

## Question 5 (40 points)

Write a program that prints an hourglass figure:

```
#include using namespace std; int main() { int n; cin >> n; if (n % 2 == 0) { cout << "Error: n must be odd" << endl; return 1; } for (int i = 0; i < n; i++) { for (int j = 0; j < n; j++) { if (i <= j && i + j < n) cout << '*'; else cout << ' '; } cout << endl; } return 0; }
```

## Question 6 (20 points)

Provide the missing constructor and methods:

```
Complex() : real(0), imaginary(0) {} Complex(double r, double i) : real(r), imaginary(i) {}  
double Complex::get_real() { return real; } double Complex::get_imaginary() { return  
imaginary; } void Complex::set_real(double r) { real = r; } void  
Complex::set_imaginary(double i) { imaginary = i; }
```